

Ionic Conductivity of Solid Electrolytes Based on Lithium Titanium Phosphate

Hiromichi Aono and Eisuke Sugimoto

Department of Industrial Chemistry, Niihama National College of Technology, Yagumo-cho 7-1, Niihama, Ehime 792, Japan

Yoshihiko Sadaoka

Department of Industrial Chemistry, Faculty of Engineering, Ehime University, Bunkyo-cho 3, Matsuyama, Ehime 790, Japan

Nobuhito Imanaka and Gin-ya Adachi

Department of Applied Chemistry, Faculty of Engineering, Osaka University, Yamadaoka 2-1, Suita, Osaka 565, Japan

ABSTRACT

Solid electrolytes based on lithium titanium phosphate were prepared, and their phase, porosity of the sintered pellets, and electrical conductivity were studied. The conductivity was increased and the porosity decreased greatly by partially replacing Ti^{4+} and P^{5+} in $\text{LiTi}_2(\text{PO}_4)_3$ with M^{3+} ($\text{M}^{3+} = \text{Al}^{3+}, \text{Cr}^{3+}, \text{Ga}^{3+}, \text{Fe}^{3+}, \text{Sc}^{3+}, \text{In}^{3+}, \text{Lu}^{3+}, \text{Y}^{3+}$, and La^{3+}) and Si^{4+} ions, respectively. The maximum conductivity at 298 K is $7 \times 10^{-4} \text{ S cm}^{-1}$ for $\text{Li}_{1.3}\text{M}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ ($\text{M} = \text{Al}$ and Sc). The conductivity was considerably increased by the mixing of binders such as Li_2O or $\text{Li}_4\text{P}_2\text{O}_7$ with $\text{LiTi}_2(\text{PO}_4)_3$. The main reason for the conductivity enhancement of these electrolytes seems to be attributable to the increase of the sintered pellet density with the enhancement of the lithium concentration at the grain boundaries.

Recently high energy density and long life batteries using solid electrolytes have been requested. Lithium ion conductors especially are promising as the electrolyte for this purpose.

Li_3N is well-known as the highest conductivity lithium conductor at room temperature (1, 2). The conductivity of glasses based on Li_2S is also reasonably high (3, 4). However, these materials are not stable in an air atmosphere. A NASICON-type network structure (5, 6) is promising in its high conductivity and stability in air. The $\text{Na}_{1+x}\text{Zr}_2\text{Si}_x\text{P}_{3-x}\text{O}_{12}$ system (NASICON) whose P^{5+} site is partially substituted with Si^{4+} in $\text{NaZr}_2(\text{PO}_4)_3$ shows a high sodium ionic conductivity. However, the $\text{Li}_{1+x}\text{Zr}_2\text{Si}_x\text{P}_{3-x}\text{O}_{12}$ system, which is a Li analogue of the $\text{Na}_{1+x}\text{Zr}_2\text{Si}_x\text{P}_{3-x}\text{O}_{12}$ system, exhibits a low ionic conductivity because a Li^+ ion is too small in size to migrate freely in their three-dimensional network structures.

The ionic conductivity for the system is greatly increased when Zr^{4+} in $\text{LiZr}_2(\text{PO}_4)_3$ is replaced by the smaller Ti^{4+} ion (7), since tunnels in $\text{LiTi}_2(\text{PO}_4)_3$ are more suitable in size for the lithium ion migration. It has been already reported that the conductivity for $\text{LiTi}_2(\text{PO}_4)_3$ increases appreciably if Ti^{4+} in the phosphate is partially substituted by a larger cation such as Ga^{3+} (8), Cr^{3+} (9), Sc^{3+} (7), or In^{3+} (10, 11). However, the reasons for the conductivity enhancement by the M^{3+} substitution are difficult to determine because of their close ionic radii. In our preceding paper (12), substitution effects of Ti^{4+} in $\text{LiTi}_2(\text{PO}_4)_3$ by various ions (Al^{3+} , Sc^{3+} , Y^{3+} , and La^{3+}) on the conductivities have been discussed. Further, the reasons for the conductivity improvement by the M^{3+} substitution are ascribed to the increase in lithium ion content and to the densification of the sample pellet adopted for the measurement.

In this paper, $\text{Li}_{1+x}\text{Ti}_2\text{Si}_x\text{P}_{3-x}\text{O}_{12}$ and $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ ($\text{M}^{3+} = \text{Al}^{3+}, \text{Cr}^{3+}, \text{Ga}^{3+}, \text{Fe}^{3+}, \text{Sc}^{3+}, \text{In}^{3+}, \text{Lu}^{3+}, \text{Y}^{3+}$, and La^{3+}) systems are prepared, and their phase, ionic conductivity, and porosity are investigated.

Experimental

Materials.—A stoichiometric mixture of Li_2CO_3 , SiO_2 , M_2O_3 ($\text{M} = \text{Al}, \text{Cr}, \text{Ga}, \text{Fe}, \text{Sc}, \text{In}, \text{Lu}, \text{Y}$, and La), TiO_2 , and $(\text{NH}_4)_2\text{HPO}_4$, whose purities were at least reagent grade, was ground and heated in a platinum crucible at 1173 K for 2h. The resulting material was reground into fine powder using a ball mill for 6h by a wet process. The dried powder was reheated at 1173 K for 2h and then ballmilled again for

12h. The particle size of the dried powder was smaller than $1 \mu\text{m}$, which was determined by a centrifugal particle size analyzer (Shimadzu SA-CP3). A suitable amount of 3% PVA solution was added to the powder as a binder, and the mixture was pressed into pellets at a pressure of $1 \times 10^8 \text{ Pa}$. The pellets were sintered at 1173 ~ 1523 K for 2h. The sintering temperature was chosen in such a way that high density (low porosity) pellets were obtained. Table I shows the sintering temperature ($x = 0.2\text{--}0.5$) and the density of the pellets ($x = 0.3$) for the $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ systems. These sintered pellets were polished with 800 grade emery paper and gold was deposited by a vacuum evaporation onto both surfaces of the disk so as to obtain a good contact between the electrolyte and the electrode.

Measurements.—X-ray diffraction analysis ($\text{CuK}\alpha$ radiation) was conducted with a Rigaku Rotaflex with high purity Si powder (99.99%) as an internal standard. Electrical conductivity was measured by means of the complex impedance method with a Vector impedance meter 4800A and LCZ meter 4276A, 4277A from Hewlett Packard Company. The pellet was heated up to 473 K once before a measurement in order to eliminate the adsorbed water. Porosity of the sintered pellets was determined by the Archimedes' method.

Results and Discussion

Crystal structure.—The mixed phosphates, $\text{A(IV) B(IV)}_2(\text{PO}_4)_3$ ($\text{A} = \text{Li}, \text{Na}, \text{K}$, etc., $\text{B} = \text{Ge}, \text{Ti}, \text{Hf}, \text{Zr}$, etc.), are composed of both BO_6 octahedra and PO_4 tetrahedral (13, 14), which are linked by corners to form a three-dimensional

Table I. Sintering temperature for the $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ systems

M^{3+} ion	Sintering temperature (K)	Porosity $x = 0.3$ (%)
Al	1253-1273	4.1
Cr	1463-1523	27.9
Ga	1173-1203	8.2
Fe	1323	4.8
Sc	1473-1483	2.0
In	1273	4.4
Lu	1173	1.7
Y	1173	0.5
La	1173	0.1

Table II. Lattice constants (hexagonal unit), the conductivities at 298 K, and porosities of the sintered pellets for the $\text{Li}_{1+x}\text{Ti}_2\text{Si}_x\text{P}_{3-x}\text{O}_{12}$ system

Sample	Lattice constants		σ (298 K) S cm^{-1}	Porosity %
	a	c		
$\text{LiTi}_2(\text{PO}_4)_3$	8.512	20.858	2.0×10^{-6}	34.0
$\text{Li}_{1.2}\text{Ti}_2\text{Si}_{0.2}\text{P}_{2.8}\text{O}_{12}$	8.518	20.848	8.6×10^{-5}	4.5
$\text{Li}_{1.3}\text{Ti}_2\text{Si}_{0.3}\text{P}_{2.7}\text{O}_{12}$	8.520	20.862	3.2×10^{-4}	3.3
$\text{Li}_{1.4}\text{Ti}_2\text{Si}_{0.4}\text{P}_{2.6}\text{O}_{12}$	—	—	1.5×10^{-4}	6.9
$\text{Li}_{1.5}\text{Ti}_2\text{Si}_{0.5}\text{P}_{2.5}\text{O}_{12}$	—	—	9.6×10^{-5}	6.2

network with the space group $R\bar{3}C$. The x-ray diffraction patterns of the solid electrolytes based on the $\text{LiTi}_2(\text{PO}_4)_3$ prepared in this study were in agreement with that of a rhombohedral ($R\bar{3}C$) structure.

$\text{Li}_{1+x}\text{Ti}_2\text{Si}_x\text{P}_{3-x}\text{O}_{12}$ system.—Lattice constants, conductivities at 298 K, and the porosities of the sintered pellets for the $\text{Li}_{1+x}\text{Ti}_2\text{Si}_x\text{P}_{3-x}\text{O}_{12}$ system are tabulated in Table II. The sintering of the samples was conducted at 1473 K for 2h. Lattice constants of the $\text{Li}_{1+x}\text{Ti}_2\text{Si}_x\text{P}_{3-x}\text{O}_{12}$ system for $x > 0.3$ could not be obtained because peaks in the x-ray patterns were not sharp enough. The cell constant for the a-axis increased slightly, indicating that the replacement of a P^{5+} ion with a Si^{4+} ion occurred. However, unknown peaks appeared in the patterns for the entire region of the $\text{Li}_{1+x}\text{Ti}_2\text{Si}_x\text{P}_{3-x}\text{O}_{12}$ system since the added Si^{4+} did not always replace at the P^{5+} site. The porosity of the sintered pellets decreased remarkably and the conductivities were greatly enhanced with the increase of Si^{4+} substitution. The maximum conductivity, $3.2 \times 10^{-4} \text{ S cm}^{-1}$ at 298 K, was obtained at $x = 0.3$.

$\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ systems.—Lattice constants vs. x value for the $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ systems are presented in Fig. 1. Limits of phase of the $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ systems determined from powder x-ray analysis are plotted against the ionic radii (15) of M^{3+} and the x value in Fig. 2. Single (rhombohedral) and mixed [$\text{LiTi}_2(\text{PO}_4)_3 + \text{Li}_3\text{M}_2(\text{PO}_4)_3$] phase areas are also shown in the same figure. When Ti^{4+} sites in

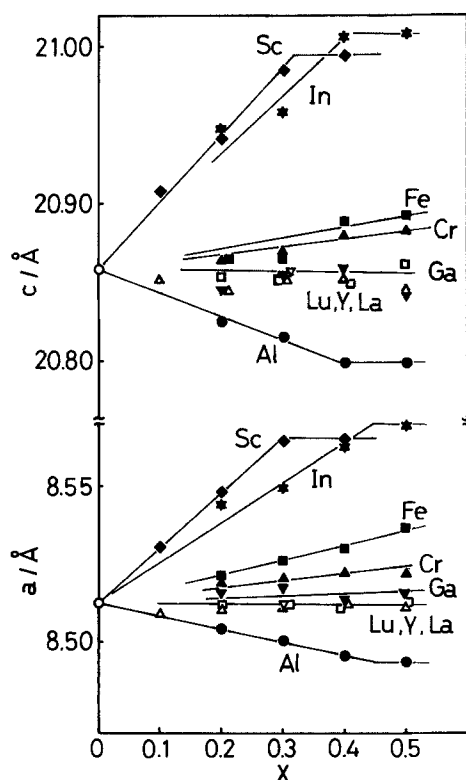


Fig. 1. Lattice constants (hexagonal unit) for the $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ systems. $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$: (●) $\text{M} = \text{Al}$, (▲) Cr , (▼) Ga , (■) Fe , (◆) Sc , (★) In , (▽) Lu , (△) Y , (□) La ; (○) $\text{LiTi}_2(\text{PO}_4)_3$ ($x = 0$).

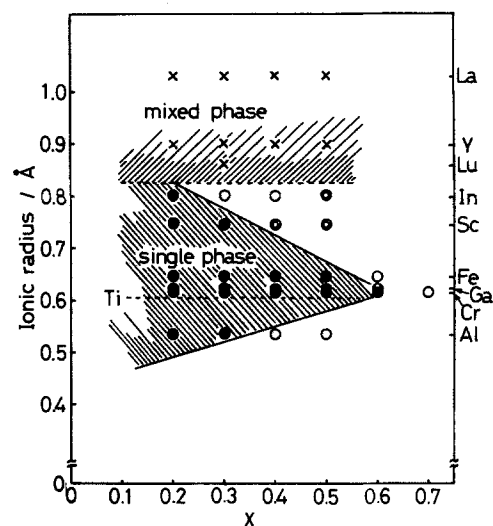


Fig. 2. Limits of phase of the $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ systems determined from x-ray analysis are plotted against the ionic radii of M^{3+} and the x value. (●) Single phase of the $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ (rhombohedral), (X) mixed phase of $\text{LiTi}_2(\text{PO}_4)_3$ and $\text{Li}_3\text{M}_2(\text{PO}_4)_3$, (○) monoclinic phase of the $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$, (○) the $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ with unknown phase.

$\text{LiTi}_2(\text{PO}_4)_3$ were partially substituted by Al^{3+} , the lattice constants decreased linearly with the increase in Al^{3+} up to $x = 0.4$ because the ionic radius of Al^{3+} was smaller than that of Ti^{4+} (Fig. 1). The single phase was formed up to $x = 0.3$, while an unknown phase appeared for $x > 0.4$ (Fig. 2). The replaceable limit for an Al^{3+} ion was around $x = 0.4$. In the substituted system for a slightly larger cation (Cr^{3+} , Ga^{3+} , or Fe^{3+}) than Ti^{4+} , the lattice constants slightly increased with the M^{3+} fraction and no second phase appeared for $x < 0.6$. The M^{3+} (Cr^{3+} , Ga^{3+} , or Fe^{3+}) ion is easy to place at the Ti^{4+} site because its ionic radius is close to that of Ti^{4+} . In the samples of the substituted systems for the larger Sc^{3+} or In^{3+} , the lattice constants increased linearly with increase of x in the $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ systems. A monoclinic phase formed as a second phase with the excess substitution of Sc^{3+} ($x > 0.3$) and In^{3+} ($x > 0.4$) ions for Ti^{4+} . On the other hand, the lattice constants for Lu^{3+} , Y^{3+} , or La^{3+} analogues were the same as that for $\text{LiTi}_2(\text{PO}_4)_3$. The replacement of the Ti^{4+} site with Lu^{3+} , Y^{3+} , or La^{3+} did not occur because of their larger ionic radii, and this resulted in mixed phases of $\text{LiTi}_2(\text{PO}_4)_3$ and $\text{Li}_3\text{M}_2(\text{PO}_4)_3$ ($\text{M} = \text{Lu}$, Y , or La). The relation between the lattice constants and the ionic radius of M^{3+} for $\text{Li}_{1.3}\text{M}(\text{III})_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ ($x = 0.3$) is presented in Fig. 3. Corresponding results for tetravalent substituted systems of $\text{Li}_{1.0}\text{M}(\text{IV})_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ ($\text{M} = \text{Ge}$ and Zr) were also obtained in this study and are shown in the same figure. In the samples of Al^{3+} , Cr^{3+} , Ga^{3+} , Fe^{3+} , or Sc^{3+} substituted systems, the lattice constants increased linearly with increasing ionic radius. The same results were also obtained for the tetravalent substituted systems. The lattice constants for the In^{3+} substitution deviated from the straight line in the figure, since In_2O_3 , used as a starting material, is easily vaporized at elevated temperatures. The lattice constants for the trivalent ($\text{M} = \text{Lu}$, Y , or La) substituted system were equal to those for $\text{LiTi}_2(\text{PO}_4)_3$. From these results, the $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ systems are classified into two groups.

1. Al^{3+} , Cr^{3+} , Ga^{3+} , Fe^{3+} , Sc^{3+} , or In^{3+} substituted systems: $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ single phase.
2. Lu^{3+} , Y^{3+} , or La^{3+} systems: $\text{LiTi}_2(\text{PO}_4)_3 + \text{Li}_3\text{M}_2(\text{PO}_4)_3$ mixed phases.

The relation between the porosity of the sintered pellets and the x value for the $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ systems is shown in Fig. 4. $\text{LiTi}_2(\text{PO}_4)_3$ ($x = 0$) could not be obtained as a dense pellet (porosity: 34%). Porosity was decreased considerably by the M^{3+} substitution for all systems examined. The porosity for Y^{3+} or La^{3+} mixed systems was particularly low, since the second phase formed may fill pores of the sintered $\text{LiTi}_2(\text{PO}_4)_3$ phase. On the other hand, the porosity for the Cr^{3+} system is higher than that of the other systems. The po-

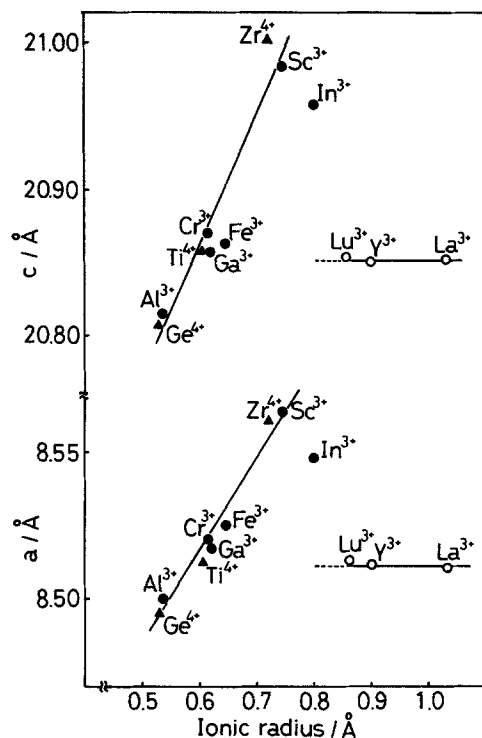


Fig. 3. Variation of the lattice constants vs. the ionic radius of the M ion. (●) $\text{Li}_{1.3}\text{M}(\text{III})_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ systems (single phase), (○) $\text{Li}_{1.3}\text{M}(\text{III})_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ systems (mixed phase), (▲) $\text{Li}_{1.0}\text{M}(\text{IV})_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ systems.

porosity variation vs. the ionic radius of M^{3+} for the $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ single-phase systems at $x = 0.2$ and 0.3 is plotted in Fig. 5. The porosity tends to have a high value when the ionic radius for the M^{3+} is close to that for Ti^{4+} . Since the ionic radius of Cr^{3+} is almost equal to that of Ti^{4+} , the porosity for the Cr^{3+} system is high, as is shown in Fig. 4. The sintering property is closely related to the ionic radius of the M^{3+} ion. We presume that all the M^{3+} ions used as a dopant do not replace the Ti^{4+} site completely, and a small amount of glassy second phase is formed at the grain boundaries. The quantity of the glassy phase increases in the case that the M^{3+} ion does not easily replace the Ti^{4+} site. Therefore, the glassy phase was not obtained for the Cr^{3+} substituted system because the easiness of the Ti^{4+} re-

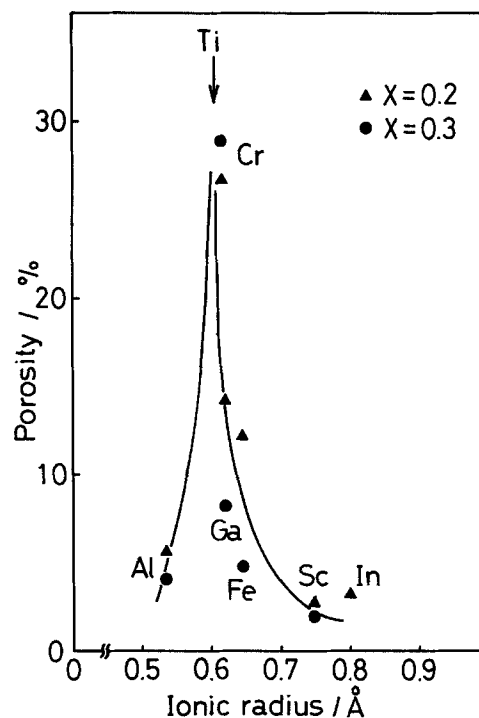


Fig. 5. The relation between the porosity and the ionic radius of M^{3+} for the $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ single-phase systems at $x = 0.2$ (▲) and $x = 0.3$ (●).

placement. The Cr^{3+} substituted phase might be analogous to the $\text{LiTi}_2(\text{PO}_4)_3$ single phase.

Figure 6 presents the conductivity results with x in the $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ systems at 298 K. The conductivity was greatly enhanced by the M^{3+} substitution and the maximum conductivity was obtained at around $x = 0.3$ for all the systems examined except for the Cr^{3+} substituted one. The conductivity for the sample of $\text{Li}_{1.3}\text{M}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ ($\text{M} = \text{Al}$ or Sc) showed the maximum value of $7 \times 10^{-4} \text{ S cm}^{-1}$ at 298 K for all the M^{3+} substitution systems. For the Al^{3+} , Cr^{3+} , Ga^{3+} , Fe^{3+} , Sc^{3+} , or In^{3+} substituted system, the conductivity could not be correlated with the ionic radius of the M^{3+} ion. In other words, the conductivity did not depend on the alteration of the cell constants by the M^{3+} substitution. The conductivity greatly increased for the Lu^{3+} , Y^{3+} , or La^{3+} group

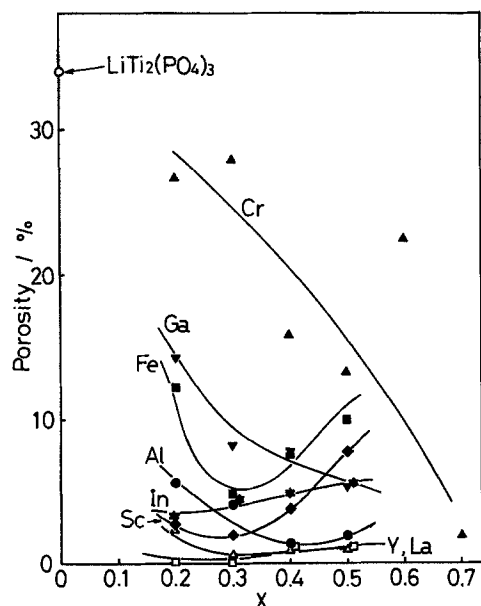


Fig. 4. Variation of the porosity for the $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ systems. $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$: (●) $\text{M} = \text{Al}$, (▲) Cr , (▼) Ga , (■) Fe , (◆) Sc , (★) In , (△) Y , (□) La ; (○) $\text{LiTi}_2(\text{PO}_4)_3$ ($x = 0$).

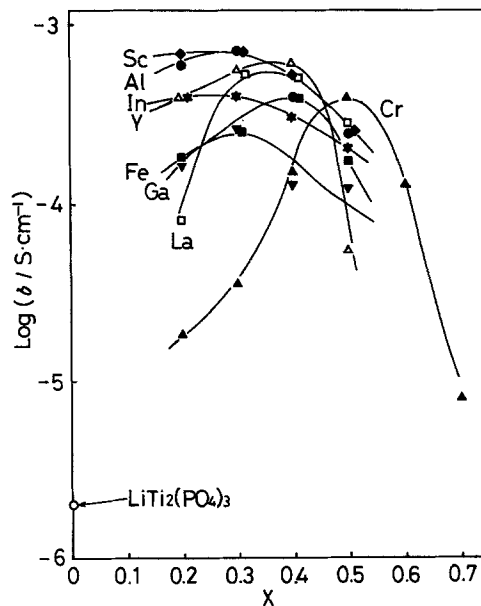


Fig. 6. Variation of the conductivity for the $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ systems at 298 K. $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$: (●) $\text{M} = \text{Al}$, (▲) Cr , (▼) Ga , (■) Fe , (◆) Sc , (★) In , (△) Y , (□) La ; (○) $\text{LiTi}_2(\text{PO}_4)_3$ ($x = 0$).

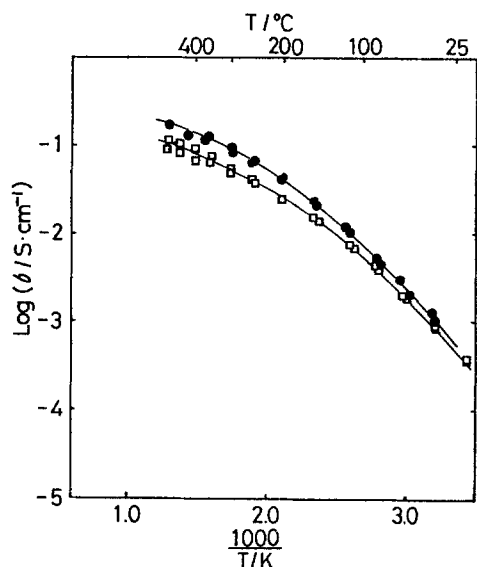


Fig. 7. The σ - $1/T$ relation for the sample of $\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ (●) and $\text{Li}_{1.3}\text{La}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ (□).

compared with the $\text{LiTi}_2(\text{PO}_4)_3$, although the Ti^{4+} site was not replaced by these ions.

The temperature dependence of conductivity for $\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ and $\text{Li}_{1.3}\text{La}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ is presented in Fig. 7. A gentle curve appeared over the entire temperature range studied, but no phase transition was observed from a DTA measurement. The activation energies at low and high (>473 K) temperatures are about 0.35 and 0.20 eV, respectively.

Figure 8 shows the relation between the conductivity at 298 K and the porosity of the sintered pellets for single phase when dopant concentrations are $x = 0.2$ and $x = 0.3$, and mixed phase ($x = 0.3, 0.4$) in Fig. 2 in addition to the $\text{LiTi}_2(\text{PO}_4)_3$ phase. The conductivity increased with the decrease of the porosity. In other words, the conductivity greatly depended on the density of the sintered pellets. The conductivity for the group of the mixed phase systems (Lu^{3+} , Y^{3+} , or La^{3+}) was smaller than that for the single-phase ones (Al^{3+} , Cr^{3+} , Ga^{3+} , Fe^{3+} , Sc^{3+} , or In^{3+}). The mixed-phase systems showed high conductivity and density in comparison with single phase $\text{LiTi}_2(\text{PO}_4)_3$. In the mixed-phase system, the conductive phase for the lithium ion is the three-dimensional network in $\text{LiTi}_2(\text{PO}_4)_3$, as the sec-

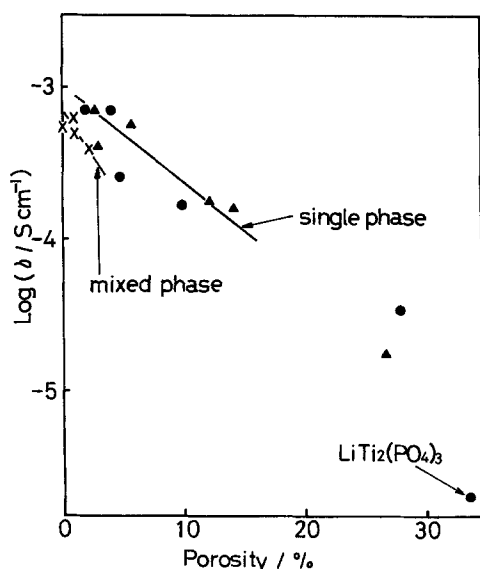


Fig. 8. The relation between the conductivity at 298 K and the porosity of the sintered pellets. The sample of the single phase ($\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$) at $x = 0.2$ (▲) and $x = 0.3$ (●). The sample of the mixed phase [$\text{LiTi}_2(\text{PO}_4)_3 + \text{Li}_3\text{M}_2(\text{PO}_4)_3$] (X).

Table III. The conductivities at 298 K and the porosities for the samples using a binder

Sample	σ (298 K) S cm^{-1}	Porosity %
$\text{LiTi}_2(\text{PO}_4)_3$	2.0×10^{-6}	34.0
$\text{LiTi}_2(\text{PO}_4)_3 + 0.2 \text{ Li}_2\text{O}$	2.4×10^{-4}	4.1
$\text{LiTi}_2(\text{PO}_4)_3 + 0.3 \text{ Li}_2\text{O}$	3.6×10^{-4}	3.0
$\text{LiTi}_2(\text{PO}_4)_3 + 0.4 \text{ Li}_2\text{O}$	1.3×10^{-4}	1.8
$\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3 + 0.3 \text{ Li}_2\text{O}$	3.5×10^{-4}	2.1
$\text{LiTi}_2(\text{PO}_4)_3 + 0.1 \text{ Li}_4\text{P}_2\text{O}_7$	1.6×10^{-4}	3.2

ond phase of $\text{Li}_3\text{M}_2(\text{PO}_4)_3$ is a lower conductivity ionic conductor, which was confirmed by our measurements in this study. The second phase is influential only on the density of the pellet. The main reason for the conductivity enhancement by the M^{3+} substitution is the increase of the sintered pellet density, which is very much affected by the existence of the second phase.

In order to clarify the effect of high densification on the conductivity increase, a stoichiometric mixture of Li_2CO_3 or $(\text{NH}_4)_2\text{HPO}_4$ was used as a binder for the starting materials of $\text{LiTi}_2(\text{PO}_4)_3$ or $\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ preparation. The binders decompose into Li_2O or $\text{Li}_4\text{P}_2\text{O}_7$ by the heat-treatment. Preparation for these powdered samples was the same as that for the $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ systems. All of these samples were sintered at 1173 K for 2 h. The sintered samples were mixed phases [$\text{LiTi}_2(\text{PO}_4)_3$ + a second phase]. However, clear peaks of a second phase were not observed from the x-ray diffraction analysis. A glassy phase might be formed in appreciable quantity by the binder utilization. Their conductivities at 298 K and the porosities are shown in Table III. The conductivity was increased and the porosity decreased greatly by the binder addition. The second-phase addition results in the increase of the lithium content, which is very effective for the enhancement of the conductivity. The lithium concentration at the grain boundaries is directly related to the lithium ion migration. The conductivity of the $\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3 + 0.3 \text{ Li}_2\text{O}$ sample is almost the same as that of the $\text{LiTi}_2(\text{PO}_4)_3 + 0.3 \text{ Li}_2\text{O}$. The Al^{3+} substitution for Ti^{4+} did not influence the conductivity increase. This result indicates that the increase of the sintered pellet density by the binder addition is the main reason for the conductivity enhancement. Table IV presents the conductivity of the bulk, the grain boundary, and the total for the samples of $\text{LiTi}_2(\text{PO}_4)_3$, $\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$, and $\text{LiTi}_2(\text{PO}_4)_3 + 0.3 \text{ Li}_2\text{O}$ at 298 K. σ_b , σ_g , and σ_{b+g} are the conductivity of the bulk, the grain boundary, and the total value, respectively. The σ_b increased by the Al^{3+} substitution more than one order of magnitude in comparison with that of the $\text{LiTi}_2(\text{PO}_4)_3$. However, the σ_b value was also enhanced by the Li_2O mixing, although the second phase is only influential on the density. For all the systems, the σ_{b+g} value is almost equal to σ_g . This means that conduction at grain boundaries is the rate-determining step for the lithium ion migration at room temperature.

Figure 9 shows the surface microstructure of the sintered pellets by a scanning electron microscopy (SEM). The $\text{LiTi}_2(\text{PO}_4)_3$ sample was porous (porosity: 34%) (Fig. 9a). Its particle size was around 1 μm , which is the same as that of the starting powder. The sintered pellet of $\text{Li}_{1.3}\text{Sc}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ was representative of the single-phase samples (porosity: 2.0%) (Fig. 9b), and the grain size is appreciably larger than that of the $\text{LiTi}_2(\text{PO}_4)_3$. The pellet is easy to sinter to high density by the replacement of the Ti^{4+} site with M^{3+} . In the case of the mixed phases system for $\text{Li}_{1.3}\text{La}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$, the grains are in close contact with each other (porosity: 0.1%) (Fig. 9c).

Table IV. The conductivity of the bulk, the grain boundaries, and the total at 298 K

Sample	σ_b (S cm^{-1})	σ_g (S cm^{-1})	σ_{b+g} (S cm^{-1})
$\text{LiTi}_2(\text{PO}_4)_3$	1×10^{-4}	2×10^{-6}	2×10^{-6}
$\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$	3×10^{-3}	9×10^{-4}	7×10^{-4}
$\text{LiTi}_2(\text{PO}_4)_3 + 0.3 \text{ Li}_2\text{O}$	1.5×10^{-3}	5×10^{-4}	3.6×10^{-4}

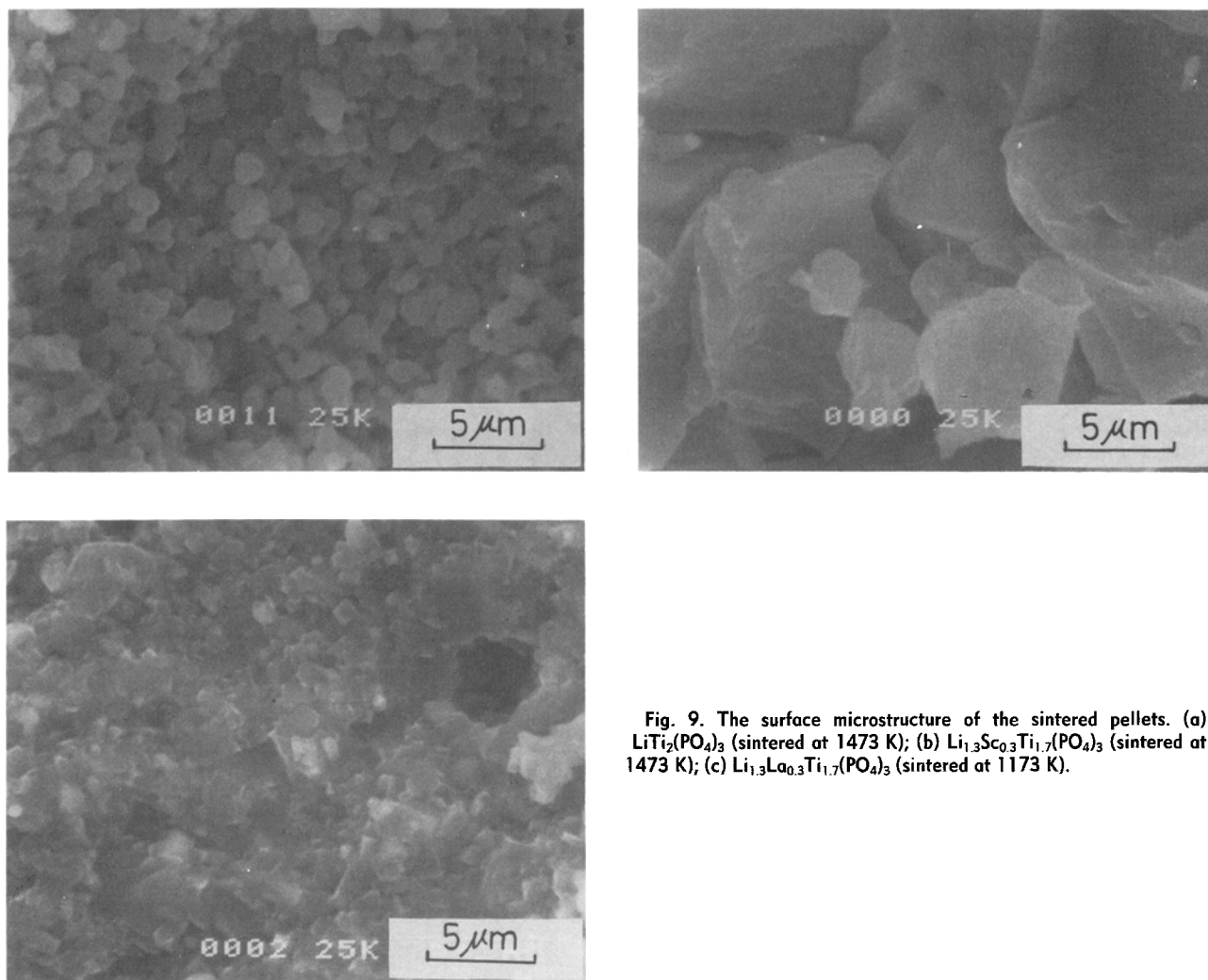


Fig. 9. The surface microstructure of the sintered pellets. (a) $\text{LiTi}_2(\text{PO}_4)_3$ (sintered at 1473 K); (b) $\text{Li}_{1.3}\text{Sc}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ (sintered at 1473 K); (c) $\text{Li}_{1.3}\text{La}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ (sintered at 1173 K).

Conclusion

$\text{LiTi}_2(\text{PO}_4)_3$ had poor ionic conductivity and showed high porosity in the sintered pellets. The partial substitution of the P^{5+} site with Si^{4+} or of the Ti^{4+} site with M^{3+} in the $\text{LiTi}_2(\text{PO}_4)_3$ system was very effective for the enhancement of the conductivity. Mixing binders such as Li_2O or $\text{Li}_4\text{P}_2\text{O}_7$ with $\text{LiTi}_2(\text{PO}_4)_3$ is also successful in obtaining a high conductivity lithium ionic conductor. The conductivity enhancement by the substitution or the binder addition resulted mainly from the densification of the sintered pellets. The increase of the lithium content at the grain boundaries is also effective in the conductivity enhancement.

The maximum conductivity at 298 K is $7 \times 10^{-4} \text{ S cm}^{-1}$ for $\text{Li}_{1.3}\text{M}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ ($\text{M} = \text{Al}$ or Sc), which is almost equal to that of Li_3N . The lithium ionic conductors based on lithium titanium phosphate are proved to be excellent solid electrolytes for their high lithium ionic conductivity even at room temperature and have a high stability in air.

Manuscript submitted March 20, 1989; revised manuscript received Sept. 23, 1989.

Osaka University assisted in meeting the publication costs of this article.

REFERENCES

1. U. v. Alpen, A. Rabenau, and G. H. Talat, *Appl. Phys. Lett.*, **30**, 621 (1977).
2. B. A. Boukamp and R. A. Huggins, *Mater. Res. Bull.*, **13**, 23 (1978).
3. G. Robert, J. P. Malugani, and A. Saida, *Solid State Ionics*, **3/4**, 311 (1981).
4. H. Wada, M. Menetrier, A. Levasseur, and P. Hagemmuller, *Mater. Res. Bull.*, **18**, 189 (1983).
5. H. Y.-P. Hong, *ibid.*, **13**, 117 (1978).
6. J. B. Goodenough, H. Y.-P. Hong, and J. A. Kafalas, *ibid.*, **11**, 203 (1976).
7. M. A. Subramanian, R. Subramanian, and A. Clearfield, *Solid State Ionics*, **18&19**, 562 (1986).
8. L. Zu-xiang, Y. Hui-jun, L. Shi-chun, and T. Shun-bao, *ibid.*, **18&19**, 549 (1986).
9. L. Zu-xiang, Y. Hui-jun, L. Shi-chun, and T. Shun-bao, *ibid.*, **31**, 91 (1988).
10. L. Shi-chun and L. Zu-xiang, *ibid.*, **9&10**, 835 (1983).
11. S. Hamdoun and D. Tranqui, *ibid.*, **18&19**, 587 (1986).
12. H. Aono, E. Sugimoto, Y. Sadaoka, N. Imanaka, and G. Adachi, *This Journal*, **136**, 590 (1989).
13. L. Hagman and P. Kierkegaard, *Acta Chem. Scand.*, **22**, 1822 (1968).
14. B. E. Taylor, A. D. English, and T. Berzins, *Mater. Res. Bull.*, **12**, 171 (1977).
15. R. D. Shannon, *Acta Cryst.*, **A32**, 751 (1976).